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A novel electricity price forecasting approach based on multi-attention feature fusion model optimized by variational mode decomposition

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ABSTRACT

Electricity price forecasting (EPF) is crucial for the optimal dispatch of energy markets. The increasing penetration of renewable energy for electricity generation has added more influencing variables to the electricity price curve, making the EPF more challenging. Therefore, this paper addresses electricity price data in energy markets with renewable energy generation and proposes an innovative Variational Mode Decomposition (VMD)-based multi-attention mechanism feature fusion model (V-MAF) for EPF. First, VMD processing reduces noise and captures multi-scale features in price and load sequences. Next, by integrating Gated Recurrent Units (GRU), Temporal Convolutional Networks (TCN), and Squeeze-and-Excitation Networks (SENet), a parallel network architecture combining SE-TCN and SE-GRU is constructed. This architecture captures local fluctuations and periodic patterns in VMD-separated multi-scale data, enhancing feature exploration and improving the model's ability to fit price variations. Finally, the output features from both networks are combined and fed into a Multi-Head Attention (MHA) along with the original features, allowing the model to focus on different parts of the input features from multiple perspectives. The innovative architecture enhances the ability to capture multi-scale features in time series and further focuses on key features through adaptive weight allocation of the attention mechanism. Experiments on the Singapore dataset and ablation studies demonstrated the effectiveness of VMD, SENet, and MHA in enhancing network performance. Multi-model comparisons showed that the V-MAF model outperformed others, providing more stable and accurate predictions. On Dataset 1, the V-MAF model achieved the Root Mean Square Error (RMSE) of 1.3168, reduced errors by 11.09% to 59.13% compared to other models such as XGBoost, ATT-CNN-LSTM, BiGRU, and VMD-Transformer.

1. Introduction

The growing energy demand has increasingly drawn attention to renewable energy technologies [1]. With the acceleration of energy markets construction and the global adoption of renewable energy, market liberalization and competition have increased, posing greater challenges in balancing electricity supply and demand, reducing costs, and enhancing renewable energy utilization [2]. The traditional standardized electricity prices set exclusively by the government are increasingly failing to reflect actual generation costs. Electricity price, as a fundamental element of the electricity market [3], is crucial for accurately predicting and optimizing operational strategies [4] and resource allocation of energy companies [5], thus reducing operating costs [6]. Accurate price forecasts also provide consumers with market trends that helping them in selecting the optimal times for purchasing electricity and achieving cost savings. Meanwhile, policymakers can adjust energy policies and regulatory measures based on electricity price forecasting (EPF), ensuring market fairness and stability. Unlike other commodities, electricity prices are influenced by multiple factors such as time cycles and electricity demand [7,8]. They exhibit characteristics such as periodicity, high volatility, seasonality, instability, and strong nonlinearity [9]. To maintain the stability of the power system,

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electricity prices need to sustain a dynamic balance between generation and consumption. Therefore, the trend towards implementing dynamic or real-time electricity pricing is becoming increasingly evident [10]. Additionally, the penetration of renewable energy generation into electricity production has increased the variables affecting electricity prices [11], making short-term price forecasting more complicated. Therefore, establishing an accurate EPF model is crucial for the efficient operation of the power system and for making well-informed market trading decisions.

EPF methods are typically categorized into two types: traditional forecasting methods and intelligent forecasting methods [12]. Traditional methods rely on historical data to predict future price trends [13]. Common approaches include Auto-Regressive Moving Average (ARMA) [14], Auto-Regressive Integrated Moving Average (ARIMA) [15], Generalized Auto-Regressive Conditional Heteroskedasticity (GARCH) [16], and others. These methods can provide stable forecasting results under certain conditions and they are known for their simple structure and low computational cost. However, they primarily address linear relationships and may fail to capture the full dynamics of the fluctuations and seasonal patterns often observed in electricity markets, making them less suitable for real-time forecasting tasks [17,18].

To address these issues, Artificial Intelligence (AI) based forecasting methods have received widespread attention in recent years. Machine Learning methods play a crucial role in the field of EPF. Classic Machine Learning algorithms such as Support Vector Regression (SVR) [19], Random Forest Regression (RFR) [20], Extreme Learning Machine (ELM) [21], and Extreme Gradient Boosting (XGBoost) have been widely applied in EPF and have shown promising results. The EPF model based-XGBoost achieves effective mapping from input features to output predictions by integrating multiple decision trees, capturing complex patterns and trends within the data [22]. It has demonstrated outstanding performance in wholesale price prediction in the Singapore electricity market. These methods achieve effective mapping from input features to output predictions, capturing complex patterns and trends in the data, thus performing well in EPF. However, these methods still have limitations in dealing with highly nonlinear and complex time-dependent data and have limited generalization capabilities [23].

In recent years, neural network-based models have achieved remarkable results in various fields, such as streamflow prediction [24], power load forecasting [25], and water quality prediction [26], thereby encouraging researchers to apply these approaches in EPF. While Convolutional Neural Network (CNN) effectively captures spatial features in data [27,28], their fixed-length kernels are ill-suited for the non-fixed periodicity of electricity price sequences, prompting researchers to explore the Recurrent Neural Network (RNN) with temporal recursion [29]. In Refs. [30] and [31], RNN variants, Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU), were utilized for EPF in the power market. Among them, GRU features a more streamlined dual-gate architecture compared to LSTM, and in Ref. [31], GRU demonstrated superior performance over LSTM. However, when processing long sequential data, both require incremental updates to the hidden state, which reduces training efficiency. To address this issue, Ref. [32] employed Temporal Convolutional Networks (TCN), which efficiently handles data, especially long time series, through a highly parallelized approach.

To further enhance prediction performance, numerous hybrid methods have been proposed, combining two or more EPF approaches rather than relying on a single prediction model. These hybrid methods aim to leverage the strengths of each individual technique [33]. Time series data often includes jumps and spikes [34], which can degrade data quality and affect prediction accuracy. In this context, Ref. [35] proposed a model combining Wavelet Transform (WT) with a hybrid LSTM, in which the dataset processed by WT decomposition exhibited more stable variance, enhancing the overall prediction accuracy of the model. However, when dealing with complex signals, methods such as WT and Empirical Mode Decomposition (EMD) may suffer from mode mixing issues, which can reduce prediction accuracy. Variational Mode Decomposition (VMD) is designed as a completely non-recursive and quasi-orthogonal method to effectively address the aforementioned shortcoming [36]. In Ref. [37], VMD was used to decompose electricity prices in California. This decomposition was combined with a Generalized Regression Neural Network (GRNN) model for prediction, demonstrating that the VMD-based model outperforms the EMD-based model in terms of performance validation. These decomposition methods transform complex price sequences into more stable sub-sequences with fewer outliers, which are then used for subsequent neural network predictions. However, the aforementioned neural networks, such as LSTM, do not fully take into account the differences in characteristics among the various sub-sequences, potentially failing to fully leverage the latent information provided by the decomposition algorithms.

The attention mechanism has offered more dynamic feature selection capabilities by extracting key features from sequential data and has been increasingly applied to time series forecasting [38]. In Ref. [39], an ensemble model based on Attention-LSTM and TCN was proposed. By embedding the attention mechanism, the LSTM hidden layer focused more on critical information, while TCN extracted long-term temporal relationships from input features. The importance of this model in feature extraction and selection was experimentally validated. As a model based on the self-attention mechanism, the Transformer effectively captures global dependencies in time series and has been widely used in time series forecasting tasks [40]. In Ref. [41], Transformer leveraged a multi-layer network structure to automatically learn complex feature representations from raw data, capturing deep patterns and long-term dependencies. Its performance was experimentally verified and found to outperform XGBoost and Attention-LSTM.

However, to address various limitations of Transformer, numerous Transformer-based variants have emerged. For instance, to tackle the issues of high computational complexity and large memory usage in Transformer models when processing long sequences [42], Ref. [43] proposed Informer, which addresses the inherent high memory complexity of self-attention by introducing a ProbSparse self-attention mechanism and distilling operations. In Ref. [44], TimesNet effectively captures local patterns in time series through periodic decomposition, addressing Transformer's limitations in modeling periodic features. In Ref. [45], TimeXer innovatively combines patch-wise self-attention and variate-wise cross-attention, enabling the standard Transformer architecture to harmonize endogenous and exogenous information, thereby improving prediction accuracy and model interpretability. These models demonstrate the strong potential of attention-based network architectures in time series forecasting.

Existing studies indicate that electricity price and load data typically contain features at different time scales. Integrating data decomposition or attention mechanisms can improve the prediction accuracy of neural networks by extracting multi-scale features and enhancing feature representation. However, even when using decomposed data for prediction, most current research relies on single-network architectures, which exhibit preference biases in extracting features across different time scales and thus fail to fully utilize these multi-scale features. Moreover, although some studies employ parallel network architectures where multiple sub-models simultaneously process data at different scales and extract corresponding features, thereby leveraging the strengths of each model, effectively fusing these sub-model features to improve prediction accuracy remains a critical challenge.

To address the aforementioned issues, such as insufficient utilization of multi-scale features and the limitations of single-model architectures, this paper proposes a VMD-based Multi-Attention Feature Fusion model (V-MAF). The main contributions of this paper are summarized as follows:

1) After validating the correlation between electricity price and load data using Pearson Correlation Coefficient (PCC) and Maximal Information Coefficient (MIC), grid search-optimized VMD

Table 1
Statistical information on the 2019 and 2020 datasets.

Data type	Year	Metrics						
		Max	Min	Avg	SD	Q1	Q2	Q3
Price ($/MWh)	2019	1354.86	−1.01	98.28	52.59	81.27	90.81	102.11
	2020	1254.04	0.00	70.01	47.45	54.03	64.63	75.79
Load (MW)	2019	7195.05	4638.75	6018.29	623.42	5462.17	6018.88	6608.38
	2020	7033.56	4615.09	5865.65	577.66	5341.87	5882.55	6389.26

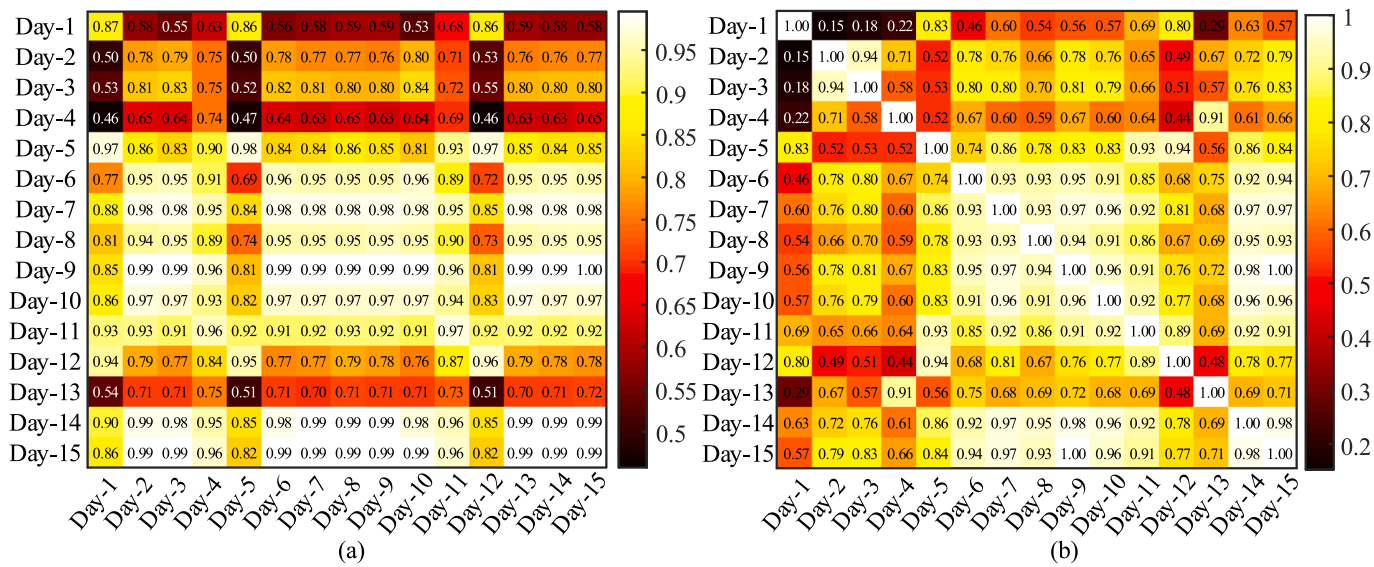

Fig. 1. PCC between past samples and current samples: (a) correlation between daily electricity prices and load, (b) correlation of daily electricity prices.

decomposes the data into multi-scale features (long-term trends, periodic components, and high-frequency fluctuations), denoising the data and reducing volatility. These features serve as network inputs, providing a basis for accurate forecasts.

2) A novel parallel GRU-TCN feature extraction module is designed to leverage their respective strengths in capturing local fluctuation features and periodic patterns from VMD-separated multi-scale data. Additionally, SENet are incorporated to enhance the extraction of critical features across different channels in each network branch.
3) A feature fusion and prediction module based on Multi-Head Attention (MHA) is designed to fully utilize the strengths of each network branch, enabling the model to focus on important features from diverse perspectives and thereby enhancing its ability to capture critical information.

The rest of this paper is organized as follows. Section 2 introduces the dataset, analyzes the correlation between electricity prices and load, and performs VMD decomposition on the electricity prices and load data. Section 3 presents the module composition and mechanisms of the proposed V-MAF method in this study. Section 4 describes the ablation and comparative experiments and provides their analysis. The conclusion is summarized in Section 5.

2. Data analysis of electricity price forecasting

In this study, the electricity prices and load data of a regional electricity market for 2019 and 2020 are selected, with a sampling interval of 30 min and 48 samples collected per day. The analysis of this dataset includes the Maximum (Max), Minimum (Min), Average (Avg), Standard Deviation (SD), and Quartiles (Q1, Q2, Q3), as detailed in Table 1.

As shown in Table 1, the electricity prices and load data for 2019 and 2020 exhibit a certain degree of dispersion in the quartile analysis. To conduct a preliminary analysis of the correlation between electricity prices and load data, the PCC is first employed to measure the linear correlation between historical electricity prices and load samples from a randomly selected half-month period and the current electricity price sample [46]. The results are presented as heat maps in Fig. 1, with the specific formula given as follows:

$$\mathrm{PCC} = \frac{\sum_{i=1}^{n}(X_i - \overline{X})(Y_i - \overline{Y})}{\sqrt{\sum_{i=1}^{n}(X_i - \overline{X})^2} \cdot \sqrt{\sum_{i=1}^{n}(Y_i - \overline{Y})^2}} \tag{1}$$

where n is the total number of samples. X_i and Y_i represent the values in the current day's data and the comparison day's data, respectively, while $\overline{X}$ and $\overline{Y}$ denote their respective means.

However, given that electricity price data and load data may exhibit significant nonlinear characteristics, the limitation of PCC in capturing only linear relationships could compromise the comprehensiveness of the analysis. Therefore, to more comprehensively capture the nonlinear dependencies between electricity price data and load data, the MIC is introduced. MIC is a statistical method based on information theory that can quantify nonlinear associations between variables, making it suitable for capturing complex nonlinear patterns. The calculation of MIC is based on Mutual Information (MI), which is a measure in information theory used to quantify the shared information between two variables. Given two variables, X and Y, representing electricity price data and load data, respectively, their MI is defined as:

$$I(X;Y) = \sum_{x \in X}\sum_{y \in Y} p(x,y)\log_2\frac{p(x,y)}{p(x)p(y)} \tag{2}$$

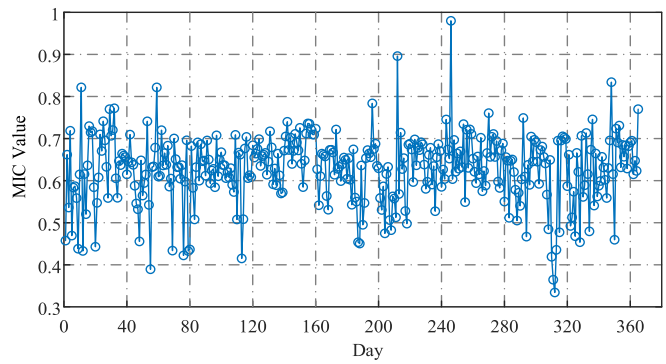

Fig. 2. Daily MIC values for electricity prices and load in 2019.

Table 2
VMD parameter optimization results through grid search.

Parameter	Tested Values	Optimal Value
K	3, 4, 5, 6, 7	5
α	1000, 1500, 2000, 2500	1500
tol	1e-7, 1e-6	1e-7

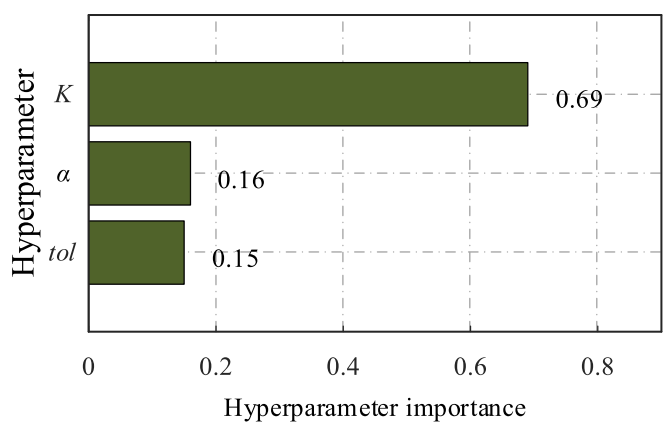

Fig. 3. Hyperparameter importance ranking in VMD.

where $p(x, y)$ is the joint probability distribution of X and Y, $p(x)$ and $p(y)$ are the marginal probability distributions of X and Y, respectively. The value of MI increases as the association between X and Y becomes stronger. To eliminate the influence of variable value ranges on MI, MIC normalizes the MI, transforming it into a value between 0 and 1, defined as:

$$MIC[X;Y] = \max_{|X||Y|<B} \frac{I(X;Y)}{\log_{2(\min(|X|,|Y|))}} \tag{3}$$

where B represents the maximum allowable number of grids, while $|X|$ and $|Y|$ denote the sample sizes of variables X and Y, respectively.

Fig. 2 illustrates the MIC values between daily electricity prices and load in the Singapore electricity market for the year 2019.

As shown in Fig. 2, the MIC values fluctuate significantly throughout the year, indicating that the non-linear correlation between load and electricity prices varies significantly over time. The overall MIC values remain relatively high, suggesting a strong non-linear dependency between load and electricity prices. The analysis of PCC and MIC demonstrates that using historical sequences of electricity prices and load for predicting current electricity prices is feasible.

Additionally, the large SD of electricity prices and loads suggests significant volatility and noise interference, as well as the presence of a certain number of outliers. Directly analyzing this data may make it difficult to effectively extract its inherent patterns. To mitigate the influence of noise and outliers, extract primary patterns and trends, and reduce the impact of price fluctuations on predictions, VMD is used to decompose the electricity prices and load signals.

VMD can adaptively decompose these complex signals into multiple Intrinsic Mode Functions (IMFs), with each IMF corresponding to oscillatory components of different frequencies and bandwidths, thereby capturing the multi-scale characteristics of the data. Compared to traditional methods, VMD effectively avoids mode mixing problem and shows strong robustness to noise and non-stationarity. By optimizing the bandwidth of each IMF, VMD more accurately extracts useful information from the signals. The main steps of VMD include defining the mode functions, constructing the variational problem, and solving the optimization problem.

The goal of VMD is to decompose a signal $y(t)$ into multiple mode functions and their corresponding center frequencies, while minimizing the bandwidth of each mode. The objective function is formulated as follows:

$$\begin{cases} \min\left\{ \sum_{k=1}^{K} \left\| \partial_t \left\{ \left[\delta(t) + j\frac{1}{\pi t} \right] * u_k(t) \right\} \cdot e^{-j\omega_k t} \right\|_2^2 \right\} \\ s.t. \sum_{k=1}^{K} u_k = y(t) \end{cases} \tag{4}$$

where u_k is the k-th mode function, ω_k is the k-th center frequency, $\delta(t)$ is the Dirac delta function, and j is the imaginary unit.

The constraint condition is introduced through the Lagrange multiplier $\eta(t)$, and the augmented Lagrangian function is constructed as follows:

$$L(u_k, \omega_k, \eta) = \alpha \sum_{k=1}^{K} \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 + \left\| y(t) - \sum_{k=1}^{K} u_k(t) \right\|_2^2 + \left\langle \eta(t), y(t) - \sum_{k=1}^{K} u_k(t) \right\rangle \tag{5}$$

where α is the quadratic penalty factor.

The alternating direction method of multipliers (ADMM) is used to solve the optimal solution of the Lagrangian function, and the variables u_k, ω_k and η are updated. The update formula is as follows:

$$\begin{cases} \hat{u}_k^{n+1}(\omega) = \dfrac{\hat{y}(\omega) - \sum_{i \neq k} \hat{u}_i(\omega) + \dfrac{\hat{\lambda}(\omega)}{2}}{1 + 2\alpha(\omega - \omega_k)^2} \\ \omega_k^{n+1} = \dfrac{\int_0^\infty \omega \left| \hat{u}_k^{n+1}(\omega) \right|^2 d\omega}{\int_0^\infty \left| \hat{u}_k^{n+1}(\omega) \right|^2 d\omega} \\ \hat{\eta}^{n+1}(\omega) = \hat{\eta}^n(\omega) + \gamma \left(\hat{y}(\omega) - \sum_{k=1}^{K} \hat{u}_k^{n+1}(\omega) \right) \end{cases} \tag{6}$$

where $\hat{u}_k^{n+1}(\omega)$ represents the $(n+1)$-th iterative value of $u_k(t)$ in the Fourier domain. $\hat{y}(\omega)$ and $\hat{\eta}^{n+1}(\omega)$ are the Fourier transforms of $y(t)$ and $\eta(t)$, respectively.

The key parameters of VMD were optimized through a systematic grid search approach, which represents a robust hyperparameter optimization methodology extensively applied in machine learning and deep learning domains. This technique operates by constructing a multidimensional grid encompassing all possible combinations of predefined hyperparameter values, followed by an exhaustive evaluation of model performance across each parameter configuration. The optimal parameter set is subsequently identified based on its validation set performance metrics. The implementation procedure comprises the

Table 3
Performance evaluation of VMD with different K values.

K values	RMSE	MAE	MAPE	R^2
3	2.2658	1.8885	2.7837	0.9369
4	1.8687	1.4327	2.0748	0.9571
5	1.3168	0.9995	1.4248	0.9787
6	1.5917	1.2898	1.8455	0.9689
7	1.8836	1.3605	1.9837	0.9564

following steps:

Initially, three critical hyperparameters were selected: the mode number (K), convergence tolerance (tol), and the data-fidelity constraint (α), with predetermined value ranges established for each parameter. Subsequently, the VMD algorithm was executed across various parameter combinations, with the resulting decomposed data serving as input for predictive model training. Model performance was quantitatively assessed through validation set evaluation, with the parameter configuration yielding the minimal Root Mean Square Error (RMSE) being identified as optimal. The comprehensive experimental outcomes are presented in Table 2.

Through systematic analysis, the relative importance of individual parameters is quantified by assessing their contribution to RMSE variations. The significance scores, which reflect the degree of influence each parameter exerts on model performance, are presented in the Fig. 3.

The parameter sensitivity analysis, presented in Fig. 3, shows that K has the greatest influence on model performance, with an importance score of 0.69, far surpassing α (0.16) and tol (0.15). Given K's dominant effect, we conducted a targeted sensitivity analysis, systematically evaluating its values from 3 to 7 while keeping the other parameters fixed (α = 1500, tol = 1e−7).

To assess the forecasting accuracy of various models, four metrics are used as evaluation indicators: Mean Absolute Error (MAE) [47], Mean Absolute Percentage Error (MAPE), RMSE [48,49], and the Coefficient of Determination (R^2). These metrics provide an objective assessment of model performance. The mathematical expressions are as follows:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|Y_{\mathrm{act}}(i) - Y_{\mathrm{pred}}(i)\right| \tag{7}$$

$$\mathrm{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{Y_{\mathrm{act}}(i) - Y_{\mathrm{pred}}(i)}{Y_{\mathrm{act}}(i)}\right| \tag{8}$$

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}\left(Y_{\mathrm{act}}(i) - Y_{\mathrm{pred}}(i)\right)^2}{n}} \tag{9}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(Y_{\mathrm{act}}(i) - Y_{\mathrm{pred}}(i)\right)^2}{\sum_{i=1}^{n}\left(Y_{\mathrm{act}}(i) - \overline{Y}\right)^2} \tag{10}$$

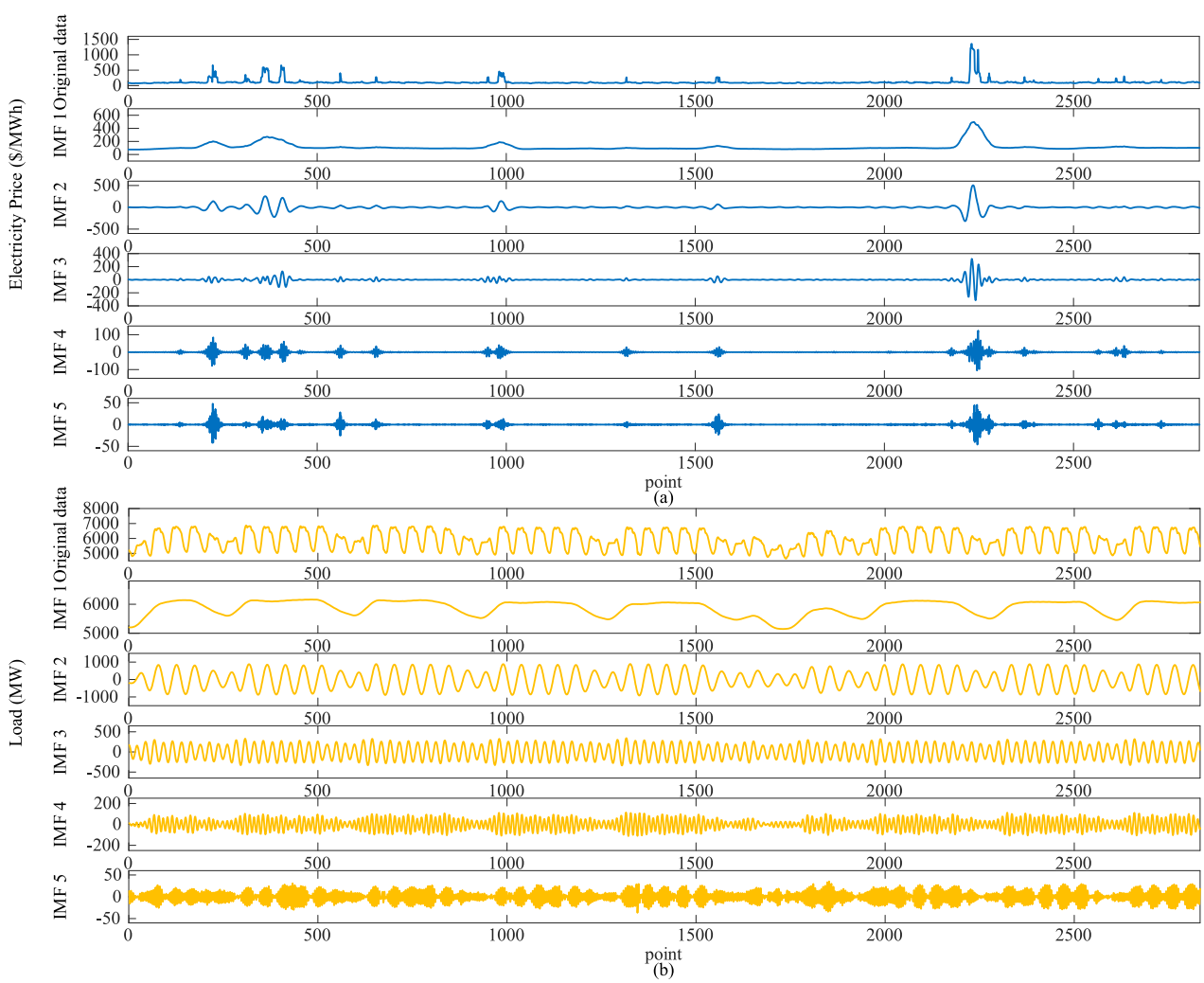

Fig. 4. Decomposition results of VMD: (a) the result of electricity price data, (b) the result of electricity load data.

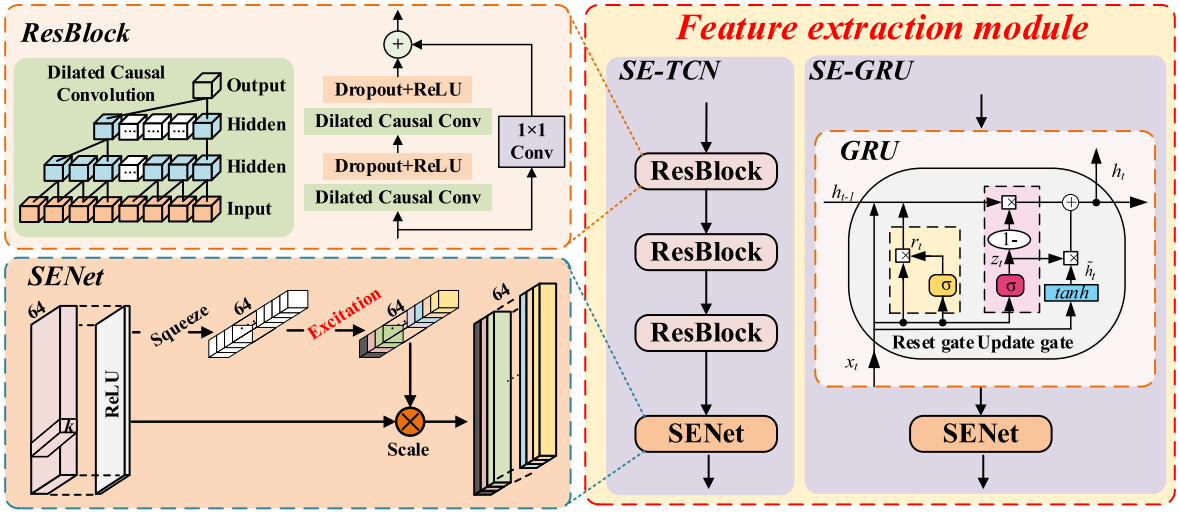

Fig. 5. The fundamental structure of feature extraction module.

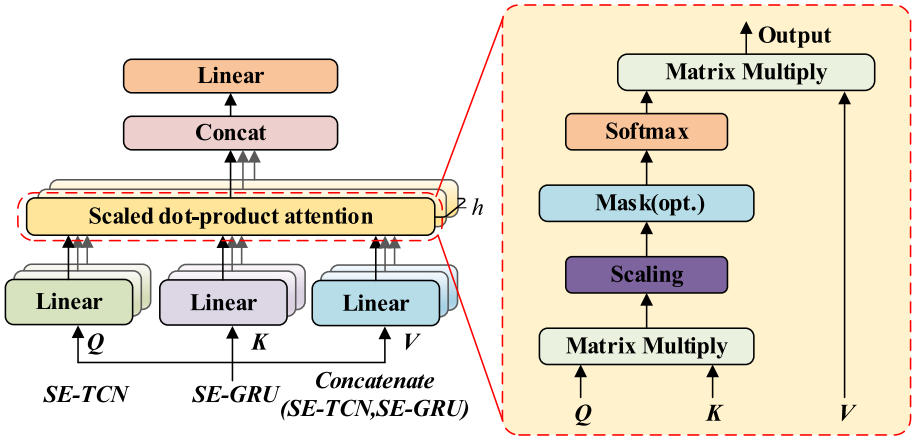

Fig. 6. Structure of the MHA.

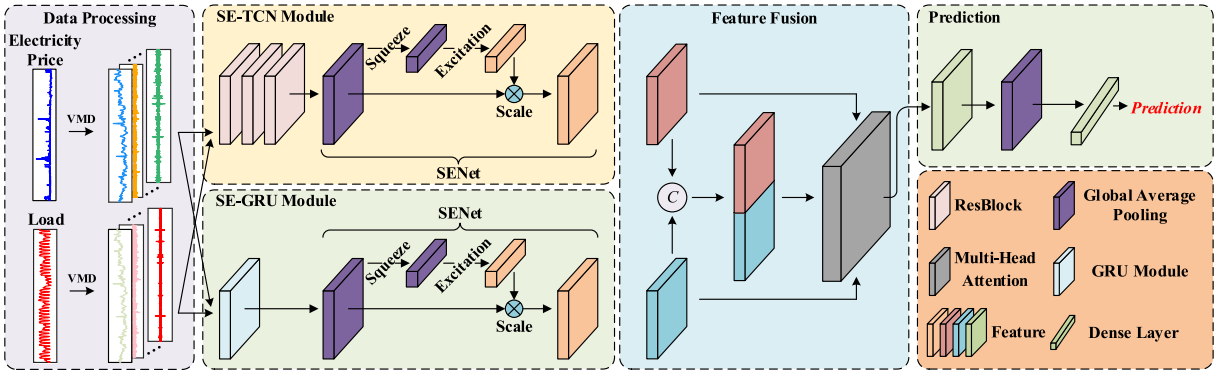

Fig. 7. Structure diagram of the V-MAF hybrid model.

where n represents the number of data samples, $Y_{\text{act}}(i)$ and $Y_{\text{pred}}(i)$ represent the true prices and the predicted value at time i, respectively, and $\overline{Y}$ represents the average of the true electricity prices.

The resulting prediction performance metrics for Dataset 1 were calculated and are shown in Table 3.

As shown by the comprehensive evaluation metrics in Table 3, the model achieves optimal performance when K is set to 5. This configuration yields the best results across all metrics, with an RMSE of 1.3168, MAE of 0.9995, MAPE of 1.4248, and an R^2 value of 0.9787. Based on these results, the parameters for VMD are determined as K = 5, tol = 1e-7, and α = 1500. Although the entire dataset has been decomposed, only the decomposition results of electricity price data and load data for January–February 2019 are presented as an example for clarity, as shown in Fig. 4.

As shown in Fig. 4, VMD has successfully decomposed the electricity price and load data into multiple IMFs, with each IMF representing

Table 4
Model performance with varying num_heads and kdim.

num_heads	kdim	RMSE	MAE	MAPE	R^2
2	16	2.1161	1.665	2.4661	0.9450
	32	1.9347	1.4361	2.1294	0.9540
	64	1.9149	1.5222	2.2187	0.9549
	128	2.0319	1.6271	2.3525	0.9493
4	16	1.7148	1.3195	1.9519	0.9639
	32	1.7531	1.301	1.9011	0.9622
	64	1.4394	1.1503	1.6752	0.9745
	128	2.0334	1.7373	2.4290	0.9492
8	16	1.6297	1.2169	1.7440	0.9674
	32	**1.3168**	**0.9995**	**1.4248**	**0.9787**
	64	1.4698	1.1417	1.6651	0.9735
	128	1.5567	1.2157	1.7825	0.9702
16	16	1.6115	1.3165	1.9255	0.9681
	32	1.4337	1.1398	1.6614	0.9747
	64	1.9368	1.5717	2.3105	0.9539
	128	2.1867	1.8486	2.5945	0.9412

Table 5
Parameters of V-MAF model structure.

Network	Layer Name	Parameters	Activation Function
SE-TCN	ResBlock-1	$f = 128$, $k = 3$, $d = 1$	ReLU
	ResBlock-2	$f = 64$, $k = 3$, $d = 2$	ReLU
	ResBlock-3	$f = 64$, $k = 3$, $d = 4$	ReLU
	SENet Block	$r = 16$	ReLU + Sigmoid
SE-GRU	GRU	units = 512	Sigmoid
	SENet Block	$r = 16$	ReLU + Sigmoid
MHA	MHA	num_heads = 8, kdim = 32	/

features at different time scales. Specifically, the low-frequency components capture long-term trends and periodic variations, while the high-frequency components reflect short-term fluctuations and local details. This separation of multi-scale features provides a clear objective for subsequent temporal feature extraction.

3. EPF framework based on V-MAF model

This section presents the specific structure of the proposed V-MAF model. The framework consists of two key modules: feature extraction and feature fusion prediction. It aims to simultaneously extract electricity prices and load features, enhance their representation, and integrate features from different models. The following will detail the implementation and workflow of these two modules within the network.

3.1. Feature extraction module

EPF is a quintessential time series prediction problem. The price of electricity is influenced by a multitude of factors, including historical prices, seasonal variations, weather conditions, and demand fluctuations. Electricity price and load data typically exhibit multi-scale temporal characteristics. VMD separates these multi-scale features, such as long-term trends, periodic patterns, and high-frequency fluctuations, by constraining the bandwidth and center frequencies, thereby assigning different frequency components to distinct IMFs. Consequently, for each component decomposed by VMD, this study employs TCN and GRU as the primary networks for extracting temporal features. As a variant of RNN, GRU, through its update and reset gates, can effectively capturing local fluctuation characteristics in electricity price data and load data, addressing short-term dependencies, and offering higher computational efficiency.

The update process of the GRU unit can be described as follows: The update gate controls the extent to which information from the previous state is incorporated into the current state, using the sigmoid function to determine z_t. The mathematical expression is as follows:

$$z_t = \sigma(\boldsymbol{W}_z[h_{t-1}, x_t] + b_z) \tag{11}$$

where $\boldsymbol{W}_z$ represents the weight matrix of the update gate, b_z represents its bias vector, and x_t represents the input value at the current time step.

The reset gate preserves only the information relevant to the current time step prediction, allowing the network to discard irrelevant information. The expression is as follows:

$$r_t = \sigma(\boldsymbol{W}_r[h_{t-1}, x_t] + b_r) \tag{12}$$

$$\widetilde{h}_t = \tanh(\boldsymbol{W}_h[r_t h_{t-1}, x_t] + b_h) \tag{13}$$

$$h_t = (1 - z_t)h_{t-1} + z_t\widetilde{h}_t \tag{14}$$

where $\boldsymbol{W}_r$ and $\boldsymbol{W}_h$ represent the weight matrices of the reset gate and the current candidate hidden state $\widetilde{h}_t$, respectively, with b_r and b_h represent their corresponding bias vectors.

However, since VMD has already separated the long-term fluctuations and periodic characteristics from the electricity price and load data, GRU may exhibit limitations in handling long-term dependencies. For instance, the vanishing gradient problem may hinder its ability to effectively capture global trends and periodic patterns. To address this limitation, this study further introduces TCN to better capture these types of features. Composed of residual connections and dilated causal convolutions, TCN provides greater flexibility in convolutional computations compared to traditional CNN, allowing for a larger receptive field in learning long sequences.

For an input sequence x, the specific output expression after feature extraction by TCN is:

$$F(s) = (x *_d f)(s) = \sum_{i=0}^{k-1} f(i) \cdot x_{s-d \cdot i}, x \leqslant 0 := 0 \tag{15}$$

where $*_d$ denotes the dilated convolution operator, f: {0, …, k-1} represents the filter, k is the size of the convolution kernel, and d is the dilation rate. The dilated convolutional kernels in TCN can cover an exponentially increasing range of the input sequence length L.

Due to the structural differences between GRU and TCN in temporal modeling, they exhibit distinct feature extraction preferences for the multi-scale components generated by VMD. These preferences can be regarded as distributed representations of multi-scale temporal features across the channel dimension. To enhance the effective features extracted by GRU and TCN, which are distributed across different channels, a Channel Attention Mechanism (CAM) is proposed to be included after the last residual block of the TCN and the output layer of the GRU. To balance structure and performance, this paper selects the SENet, a widely validated and relatively simple structure, as the implementation of the CAM.

SENet, as an attention mechanism, enhances feature extraction by modeling the interdependencies between feature channels to highlight important features and achieve adaptive weighting of features. The core idea is to use global information to learn channel weights, and its output can be expressed as follows:

$$X'_{c,i,j} = \sigma\left(\boldsymbol{W}_2 \cdot \delta\left(\boldsymbol{W}_1 \cdot \frac{1}{L \times B} \sum_{i=1}^{L} \sum_{j=1}^{B} X_{c,i,j}\right)\right) \cdot X_{c,i,j} \qquad \boldsymbol{W}_1 \in \mathbb{R}^{\frac{C}{r} \times C}, \boldsymbol{W}_2 \in \mathbb{R}^{C \times \frac{C}{r}} \tag{16}$$

where $\boldsymbol{W}_1$ and $\boldsymbol{W}_2$ are the weight matrices of the two fully connected

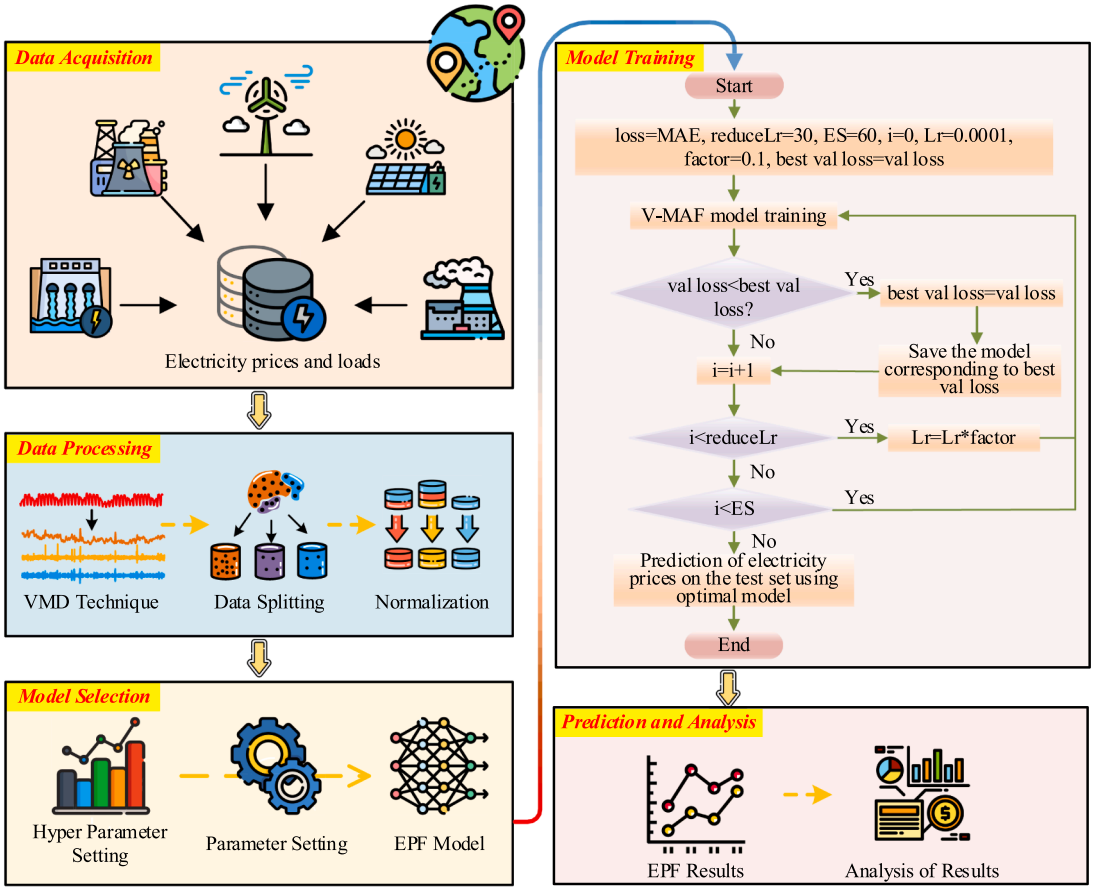

Fig. 8. The flowchart of the EPF method.

Table 6
The parameters of system configuration.

Hardware	Model	Software	Version
OS	Win10 (64bit)	Python	3.6
CPU	Intel Core i9-14900 K	TensorFlow	2.6.0
	@3.3.0 GHz	Keras	2.6.0
GPU	NVIDIA GeForce RTX 4090	Scikit-learn	1.0.2
RAM	DDR4 24 GB	CUDA	11.2.0
HDD	SSD 1 TB	cuDNN	8.1.0

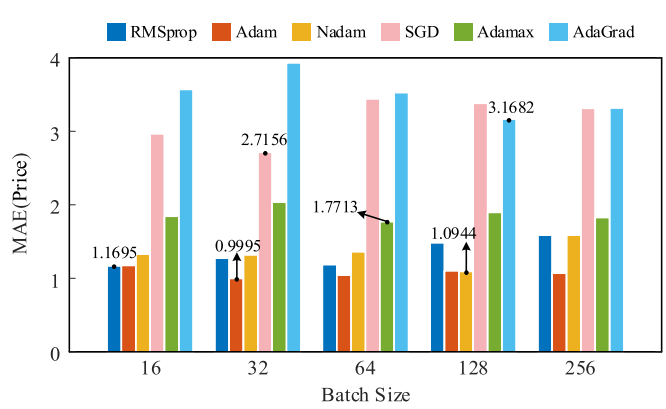

Fig. 9. MAE results for different optimizers with different batch sizes.

layers, respectively, r is the reduction ratio, $\sigma(\cdot)$ represents the sigmoid, $\delta(\cdot)$ represents the ReLU, i and j are spatial indexes.

In the feature extraction module, the parallel architecture design of TCN and GRU fully leverages the unique strengths of both networks, collectively enhancing the model's ability to capture temporal features in electricity price sequences. GRU, through its gating mechanisms, effectively models short-term dependencies in electricity price data, capturing local patterns and dynamic changes. On the other hand, TCN utilizes its dilated convolutions and residual connections to capture long-term dependencies, thereby better modeling global trends and periodic patterns. The outputs of GRU and TCN are further optimized using SENet, which dynamically adjusts feature responses across channels, enhancing the representation of important features while suppressing irrelevant ones. This feature extraction module fully exploits the advantages of both convolutional and recurrent networks, effectively extracting multi-dimensional information from temporal data, and improves the perception of key features through the attention mechanism. The structure of the feature extraction module is illustrated in Fig. 5.

3.2. Feature fusion prediction module

In the feature extraction module proposed in this study for EPF, we employ parallel SE-TCN and SE-GRU to extract temporal features separately. Subsequently, since EPF involves complex relationships and dependencies among the aforementioned multi-scale temporal features, the outputs of SE-TCN and SE-GRU are concatenated to generate a comprehensive feature representation, fully integrating the strengths of both models. The fused features, along with the original features, are then fed into a MHA mechanism. By computing multiple attention heads

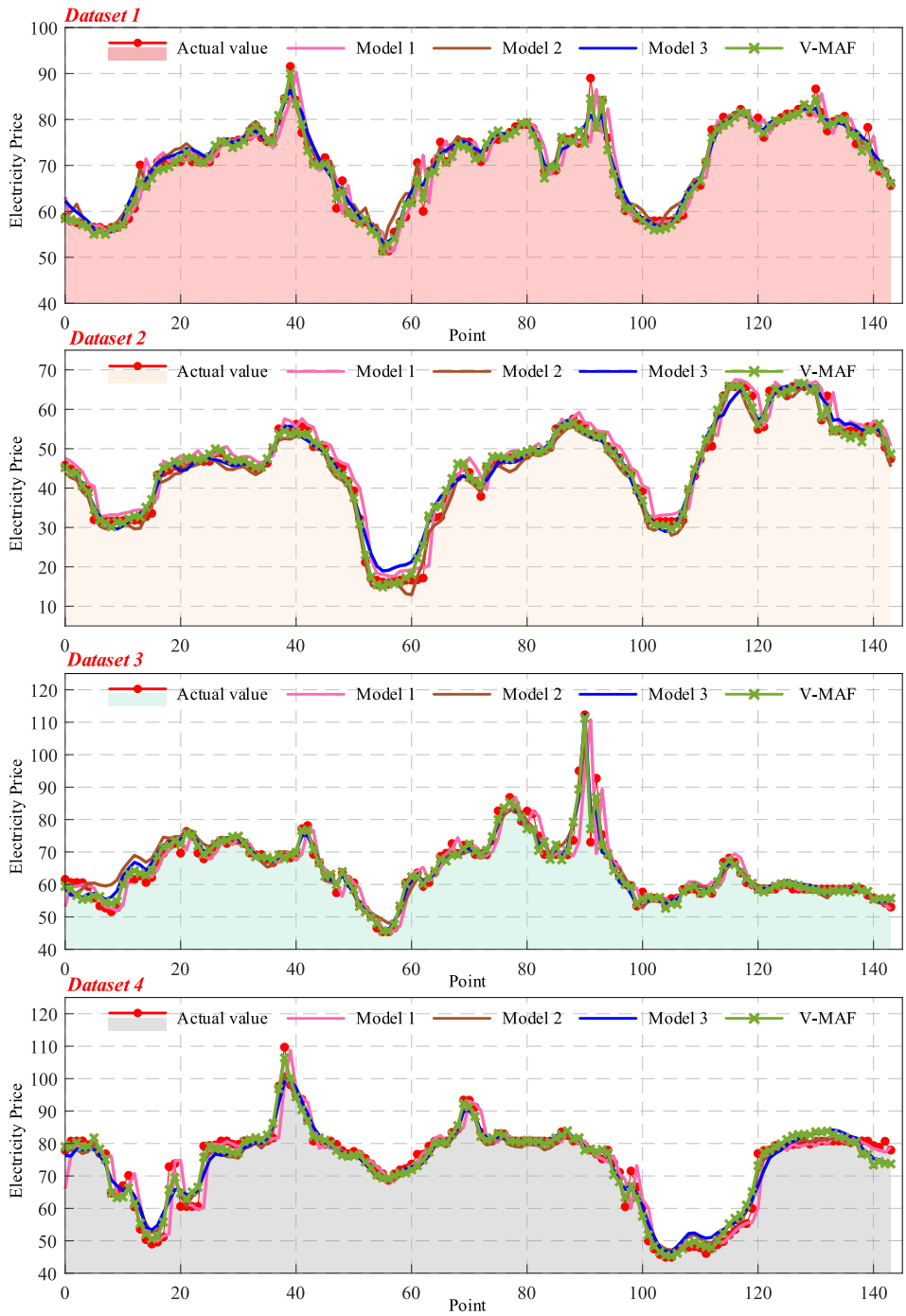

Fig. 10. Comparison of different model predictions in ablation experiments.

in parallel, the MHA mechanism assigns different weights to the input information, enabling the model to focus on different parts of the input features from various perspectives and prioritize important features, thereby enhancing its ability to capture critical information.

The attention mechanism consists of Query (Q), Key (K), and Value (V) vectors, essentially establishing a mapping relationship between the Q vector and a series of key-value pairs. The structure is shown in Fig. 6.

First, the input values are projected into Q, K, and V through linear transformations, with each head having its own projection matrices. Then, the attention output for each head is calculated separately using scaled dot-product attention:

$$\text{head}_i = f_{\text{ATT}}(\boldsymbol{Q}_i,\ \boldsymbol{K}_i,\ \boldsymbol{V}_i) \ = \ \text{Softmax}\left(\frac{\boldsymbol{Q}_i \cdot \boldsymbol{K}_i^{\text{T}}}{\sqrt{d_k}}\right)\boldsymbol{V}_i \tag{17}$$

Multi-head concatenation is performed by concatenating the output values of each head and then linearly projecting them again to obtain the final value:

$$f_{\text{MHA}}(\boldsymbol{Q},\boldsymbol{K},\boldsymbol{V}) = \text{Concat}(\text{head}_1, \text{head}_2, \ldots, \text{head}_h,)\boldsymbol{W}^O \quad \boldsymbol{W}^O \in \mathbb{R}^{hd_k \times d_{\text{model}}} \tag{18}$$

The MHA mechanism improves the model's sensitivity to electricity prices trends and prediction accuracy by computing multiple independent attention heads in parallel, each focusing on different parts of the input features from various perspectives. This enables the model to more

Table 7
Evaluation metrics for each model in the ablation experiment.

Dataset	Prediction model	Metrics			
		RMSE	MAE	MAPE	R^2
Dataset 1	Model 1	3.1338	2.0037	2.7912	0.8793
	Model 2	1.7118	1.3187	1.9768	0.9640
	Model 3	1.9513	1.4245	2.0613	0.9532
	V-MAF	**1.3168**	**0.9995**	**1.4248**	**0.9787**
Dataset 2	Model 1	3.2417	2.4493	6.3709	0.9327
	Model 2	1.9623	1.6209	4.1298	0.9753
	Model 3	2.2789	1.6391	5.0144	0.9667
	V-MAF	**1.5533**	**1.1329**	**3.074**	**0.9845**
Dataset 3	Model 1	5.2929	2.661	3.8395	0.7076
	Model 2	3.1352	2.3077	3.5698	0.8974
	Model 3	2.3670	1.6312	2.4873	0.9415
	V-MAF	**1.8466**	**1.3459**	**2.0418**	**0.9644**
Dataset 4	Model 1	4.5374	2.3509	3.2608	0.8755
	Model 2	2.7339	2.2534	3.2458	0.9548
	Model 3	3.2991	2.5531	3.7339	0.9342
	V-MAF	**2.3890**	**1.8396**	**2.6208**	**0.9655**

comprehensively understand and utilize the multi-dimensional information of the input data, significantly enhancing the accuracy and robustness of EPF.

3.3. V-MAF hybrid model framework

Given the time dependency and rich features of electricity prices and load data, the overall framework of the EPF model is illustrated in Fig. 7.

Firstly, electricity price and load data typically contain multiple time-scale characteristics. Through VMD, multi-scale features such as long-term trends, periodicity, and high-frequency fluctuations can be separated. Therefore, for each component decomposed by VMD, the aforementioned SE-TCN and SE-GRU networks are designed for parallel feature extraction. In the feature fusion prediction module, the two output features from the feature extraction module are fused using the concatenate operation to generate a comprehensive feature representation. Subsequently, these three feature representations are input into a MHA mechanism to better capture the complex relationships and interactions between different time-scale features. Finally, predictions are made through a dense layer.

In this study, the key hyperparameters of the MHA mechanism, namely num_heads and kdim, have a significant impact on model performance. To determine the optimal values for these hyperparameters, we employed the Grid Search method to systematically explore possible values for num_heads and kdim. Specifically, the range of num_heads was set to [2,4,8,16], and the range of kdim was set to [16, 32, 64, 128]. Experimental results indicate that the optimal hyperparameter configuration is num_heads = 8 and kdim = 32.

To further analyze the impact of num_heads and kdim on the model's predictive performance, Table 4 presents the performance variations of the model under different hyperparameter settings. For clarity, the optimal values of each indicator are highlighted in bold.

The experimental results in Table 4 demonstrate that the model achieves optimal performance when configured with num_heads = 8 and kdim = 32. Under this configuration, the model exhibits the best evaluation metrics, including RMSE (1.3168), MAE (0.9995), and MAPE (1.4248), which are the lowest among all tested models. Additionally, the model achieves the highest R_2 value of 0.9787, outperforming other configurations. The analysis reveals that when num_heads is fixed at 8, increasing kdim from 16 to 32 significantly enhances the model's feature representation capability, thereby reducing prediction errors. However, further increasing kdim to 64 and 128 does not yield additional performance improvements. Instead, it leads to a decline in model performance due to increased computational complexity and potential overfitting, which in turn raises prediction errors. Regarding the num_heads parameter, the experimental data indicate that increasing its value from 2 to 8 while maintaining kdim = 32 improves the model's flexibility in processing the attention mechanism. This configuration enables the model to capture diverse features of the input data from multiple perspectives, thereby enhancing overall performance. However, when num_heads is increased to 16, the model's performance begins to degrade, suggesting that an excessive number of attention heads may introduce computational redundancy and negatively impact model effectiveness. These findings indicate that the optimal balance between model complexity and performance is achieved at num_heads = 8 and kdim = 32.

The network parameters of the V-MAF hybrid model are shown in Table 5, where k represents the size of the convolution kernel, f represents the filters, d represents the dilation rate, and r is the reduction ratio.

The flowchart of the V-MAF method for EPF in this paper is shown in Fig. 8, which comprises five steps: 1) Obtaining the data of electricity prices and load in the electricity market containing renewable energy sources; 2) pre-processing the dataset, including VMD decomposition, dataset partitioning, and data normalization; 3) comparing the hyper-parameters of the V-MAF model and setting up the V-MAF model; 4) training and predicting with the model; 5) evaluating the model's predictive performance and comparing and analyzing the prediction results.

4. Experiment analysis and discussion

To evaluate the effectiveness of the V-MAF model for EPF, this section presents experiments and analyzes the results. The selected 2019 dataset is used as the training set, and the four quarters of 2020 are forecasted respectively. The construction and training of the proposed V-MAF model and the comparison model are conducted in the same training environment and dataset. In this study, Keras is used for all implementations, and the configuration of the system environment is shown in Table 6.

4.1. Selection of optimizers and batch sizes

The choice of hyper-parameters has a significant impact on the accuracy of the prediction model. To explore the appropriate selection of hyper-parameters, this study uses MAE as the loss function and introduces a learning rate decay strategy during training. When the loss rate on the validation set shows no improvement within 30 epochs, the learning rate is reduced to 10 % of its original value. Additionally, an early stopping mechanism is introduced, halting the training if there is no improvement within 60 epochs. We compared the prediction performance of the model using 6 optimizers and 5 batch sizes. Fig. 9 shows the MAE values of the prediction results under different optimizers and batch sizes, as well as the best MAE values for each optimizer, highlighting the optimal MAE value for each optimizer.

Fig. 9 shows that SGD and AdaGrad have high MAE values across all batch sizes, indicating poor performance. AdaMax performs relatively steadily, but is not as effective as Nadam and RMSprop. RMSprop performs best with a batch size of 16, achieving an MAE of 1.1695, while Nadam performs best with a batch size of 128, achieving an MAE of 1.0944. Overall, Adam maintains high accuracy across different batch sizes, particularly achieving the lowest MAE of 0.9995 with a batch size of 32. Therefore, Adam is selected as the optimizer, with a batch size of 32.

4.2. Results of prediction

This section will conduct ablation and comparative experiments.

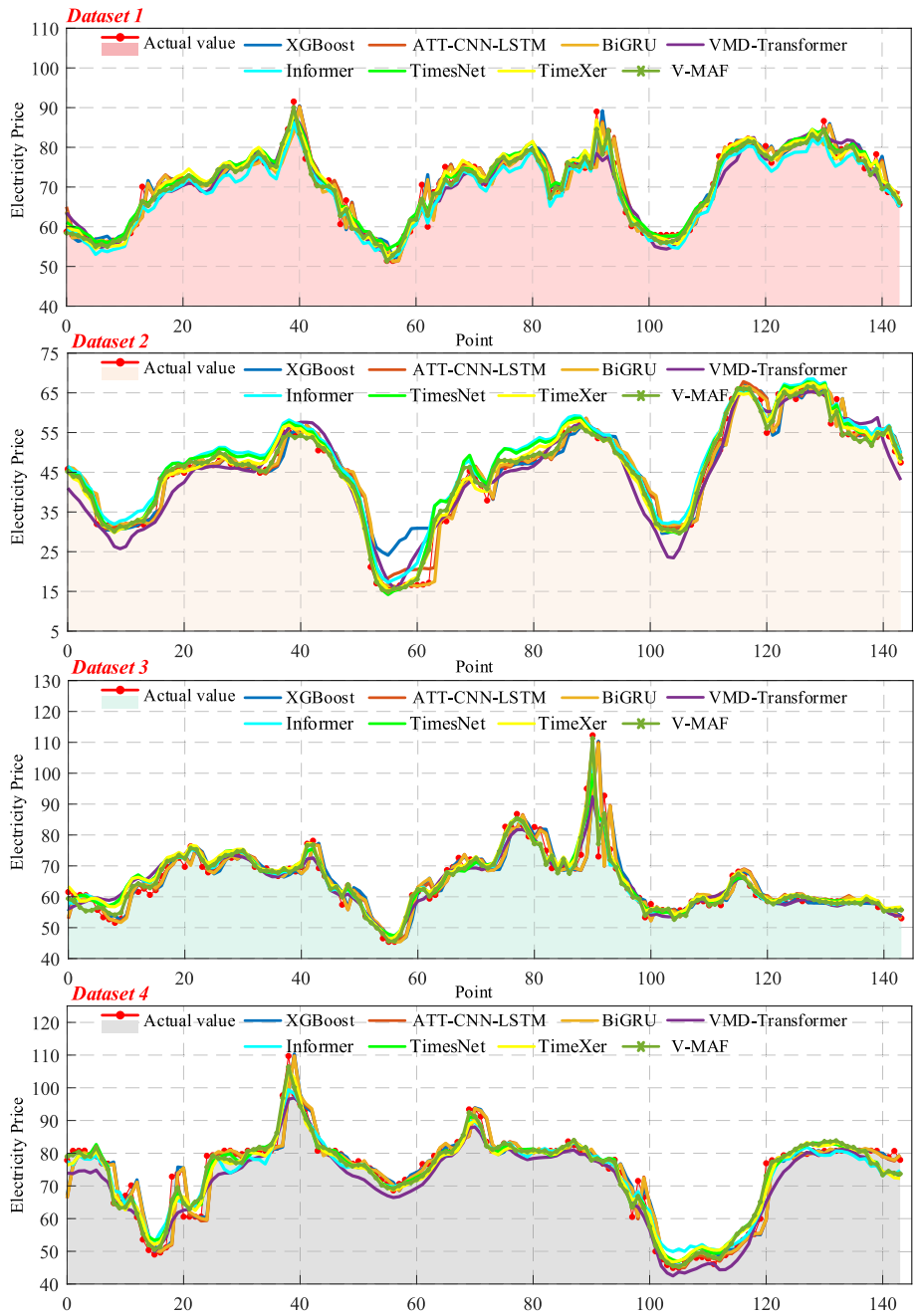

Fig. 11. Comparison of different model predictions in comparative experiment.

First, to clearly determine the effectiveness and relative importance of each part of the hybrid model, and to evaluate the training effects and contributions of various modules, we constructed four models in the ablation experiment: a model without VMD (Model 1), a model without SENet optimization (Model 2), a model without MHA optimization (Model 3), and the complete V-MAF hybrid model. We selected four quarters, specifically Dataset 1 (March), Dataset 2 (June), Dataset 3 (September), and Dataset 4 (December), and performed predictions for three days in each dataset. The forecasted prices are compared with the actual results, as shown in Fig. 10.

Fig. 10 shows that the overall trend of the electricity prices predicted by each model in the ablation experiment closely matches the actual values. To make a quantitative comparison, key performance indicators RMSE, MAE, MAPE, and R^2 were calculated. For clarity, the optimal values of each indicator for each dataset are highlighted in bold in Table 7.

As seen in Table 7, the proposed V-MAF hybrid model demonstrates significant improvements across the four datasets. In evaluating the prediction results of the V-MAF model against its variants (Model 1, Model 2, Model 3), which exclude specific modules, the V-MAF model consistently outperforms the variants on all datasets, indicating its superior overall performance. The removal of the VMD module (Model 1) has the most substantial negative impact on performance, while the removal of SENet (Model 2) and MHA (Model 3) has a relatively smaller yet still noticeable impact, making the full V-MAF model the best performer.

For example, in Dataset 1, the RMSE of the V-MAF model is 1.3168, which is an improvement of 57.98 %, 23.08 %, and 32.52 % over Model 1 (3.1338), Model 2 (1.7118), and Model 3 (1.9513), respectively. The MAE of the V-MAF model is 0.9995, improving by 50.12 %, 24.21 %, and 29.84 % compared to Model 1 (2.0037), Model 2 (1.3187), and Model 3 (1.4245). Similarly, its MAPE is 1.4248, showing improvements

Table 8
Evaluation metrics for each model in the comparison experiment.

Dataset	Prediction model	Metrics			
		RMSE	MAE	MAPE	R^2
Dataset 1	XGBoost	3.2227	2.1677	3.0594	0.8724
	ATT-CNN-LSTM	3.1922	2.1253	2.9899	0.8748
	BiGRU	3.1760	2.0495	2.8549	0.9259
	VMD-Transformer	2.4560	1.8312	2.6163	0.8844
	Informer	2.2840	1.9406	2.7298	0.9359
	TimesNet	1.7023	1.3147	1.9051	0.9644
	TimeXer	1.4810	1.1854	1.7325	0.9730
	V-MAF	**1.3168**	**0.9995**	**1.4248**	**0.9787**
Dataset 2	XGBoost	4.1234	2.4820	8.6622	0.8911
	ATT-CNN-LSTM	2.9191	1.8835	5.2199	0.9454
	BiGRU	3.0074	1.8076	4.3658	0.9421
	VMD-Transformer	3.5203	2.7391	7.6850	0.9206
	Informer	3.0754	2.4930	6.8624	0.9394
	TimesNet	2.4441	1.9181	4.9791	0.9617
	TimeXer	1.8148	1.4259	3.6757	0.9789
	V-MAF	**1.5533**	**1.1329**	**3.0740**	**0.9845**
Dataset 3	XGBoost	5.2496	2.7582	3.9867	0.7124
	ATT-CNN-LSTM	5.2262	2.6096	3.7579	0.7149
	BiGRU	5.2334	2.6270	3.7773	0.7141
	VMD-Transformer	3.3993	2.1990	3.3026	0.8794
	Informer	2.5880	1.7030	2.5825	0.9301
	TimesNet	2.7830	1.8466	2.7736	0.9192
	TimeXer	2.2493	1.6675	2.5806	0.9472
	V-MAF	**1.8466**	**1.3459**	**2.0418**	**0.9644**
Dataset 4	XGBoost	4.5523	2.3836	3.3487	0.8746
	ATT-CNN-LSTM	4.5712	2.3391	3.2381	0.8736
	BiGRU	4.5701	2.3663	3.2937	0.8737
	VMD-Transformer	4.2960	3.4132	4.7200	0.8884
	Informer	3.2208	2.4614	3.6428	0.9372
	TimesNet	2.5992	2.0388	3.0135	0.9591
	TimeXer	3.0063	2.3570	3.4122	0.9453
	V-MAF	**2.3890**	**1.8396**	**2.6208**	**0.9655**

of 48.95 %, 27.92 %, and 30.88 % over Model 1 (2.7912), Model 2 (1.9768), and Model 3 (2.0613). Finally, the R^2 of the V-MAF model is 0.9787, representing an improvement of 11.30 %, 1.52 %, and 2.68 % over Model 1 (0.8793), Model 2 (0.9640), and Model 3 (0.9532).

The ablation experiments validate the effectiveness of each module (VMD, SENet, MHA), with VMD contributing the most significantly. Specifically, the VMD module plays a critical role in decomposing the input time series into IMFs, enhancing the model's ability to capture multi-scale features. Without VMD, the model struggles to separate high-frequency noise from low-frequency trends, leading to a significant increase in prediction error, as evidenced by the sharp decline of Model 1′s metrics. The SENet module optimizes feature representation by adaptively recalibrating channel-wise responses. Its removal (Model 2) results in less discriminative feature maps, especially when certain channels contain more relevant information, which causes a moderate increase in prediction error due to the model's reduced ability to prioritize important features. The MHA module captures long-range dependencies and interactions between time steps. Without MHA (Model 3), the model's capacity to integrate global contextual information is weakened, resulting in higher prediction error, especially in datasets with complex temporal dependencies where relationships between distant time steps are critical.

In summary, the VMD, SENet, and MHA modules each play distinct yet complementary roles in the model's feature extraction, representation optimization, and temporal dependency modeling, respectively. Their combined contributions are essential for the superior performance of the full V-MAF model. Removing any of these modules leads to a notable decline in model performance.

To further verify the effectiveness of the V-MAF model in EPF and determine the performance advantages of this hybrid model compared to other existing models, we designed comparative experiments and constructed four comparison models: XGBoost, ATT-CNN-LSTM, BiGRU, VMD-Transformer, Informer, TimesNet and TimeXer. These were compared with the proposed V-MAF hybrid model. In the experiments, we applied these models to the same tasks and evaluated their performance by comparing the electricity prices prediction results of each model with the actual results across different datasets. Among them, the parameters of the XGBoost, ATT-CNN-LSTM, BiGRU, and VMD-Transformer are determined using grid search. The parameters for Informer and TimesNet are set according to Ref. [50], while the parameters for TimeXer are based on the publicly available parameters provided by the authors in Ref. [45]. The results are shown in Fig. 11.

As seen in Fig. 11, the prediction effects of each model are clearly displayed. In the comparative experiments, the overall trend of the EPF by each model closely matches the actual values. For quantitative comparison, the RMSE, MAE, MAPE, and R^2 metrics were calculated for each model, as presented in Table 8. For clarity, the optimal values of each indicator for each dataset are highlighted in bold in the table.

As shown in Table 8, when comprehensively evaluating the performance of all models across the four datasets, V-MAF emerges as the best-performing model overall, achieving the lowest RMSE, MAE, and MAPE values and the highest R^2, demonstrating strong applicability across diverse datasets. TimeXer and TimesNet are suboptimal choices, slightly inferior to V-MAF but still highly competitive. Informer and VMD-Transformer exhibit relatively stable performance but underperform compared to the aforementioned models. In contrast, XGBoost, ATT-CNN-LSTM, and BiGRU show comparatively weaker results.

Furthermore, taking Dataset 1 as an example, the percentage improvements of each metric in V-MAF relative to other models are calculated and presented in Table 9.

As shown in Tables 8 and 9, the V-MAF model demonstrates significant improvements in RMSE, MAE, MAPE, and R^2 compared to other models on Dataset 1. Specifically, the RMSE of the V-MAF model is 1.3168, with reductions ranging from 11.09 % to 59.13 % compared to XGBoost (3.2227), ATT-CNN-LSTM (3.1922), BiGRU (3.1760), VMD-Transformer (2.4560), Informer (2.2840), TimesNet (1.7023), and TimeXer (1.4810). Additionally, the MAE of the V-MAF model is 0.9995, with reductions ranging from 15.68 % to 53.90 %, while its MAPE is 1.4248, with reductions ranging from 17.76 % to 53.42 %. Furthermore, the R^2 value of the V-MAF model is 0.9787, achieving improvements of 0.58 % to 12.18 % over all other models.

Therefore, the V-MAF model demonstrates a more significant performance improvement compared to traditional models such as

Table 9
Performance enhancement of V-MAF model on Dataset 1.

Prediction model	RMSE reduction percentage	MAE reduction percentage	MAPE reduction percentage	R^2 improvement percentage
XGBoost	59.13 %	53.90 %	53.42 %	12.18 %
ATT-CNN-LSTM	58.75 %	52.97 %	52.34 %	11.88 %
BiGRU	58.53 %	51.23 %	50.10 %	5.70 %
VMD-Transformer	46.36 %	45.42 %	45.53 %	10.65 %
Informer	42.34 %	48.50 %	47.80 %	4.57 %
TimesNet	22.65 %	23.98 %	25.21 %	1.48 %
TimeXer	11.09 %	15.68 %	17.76 %	0.58 %

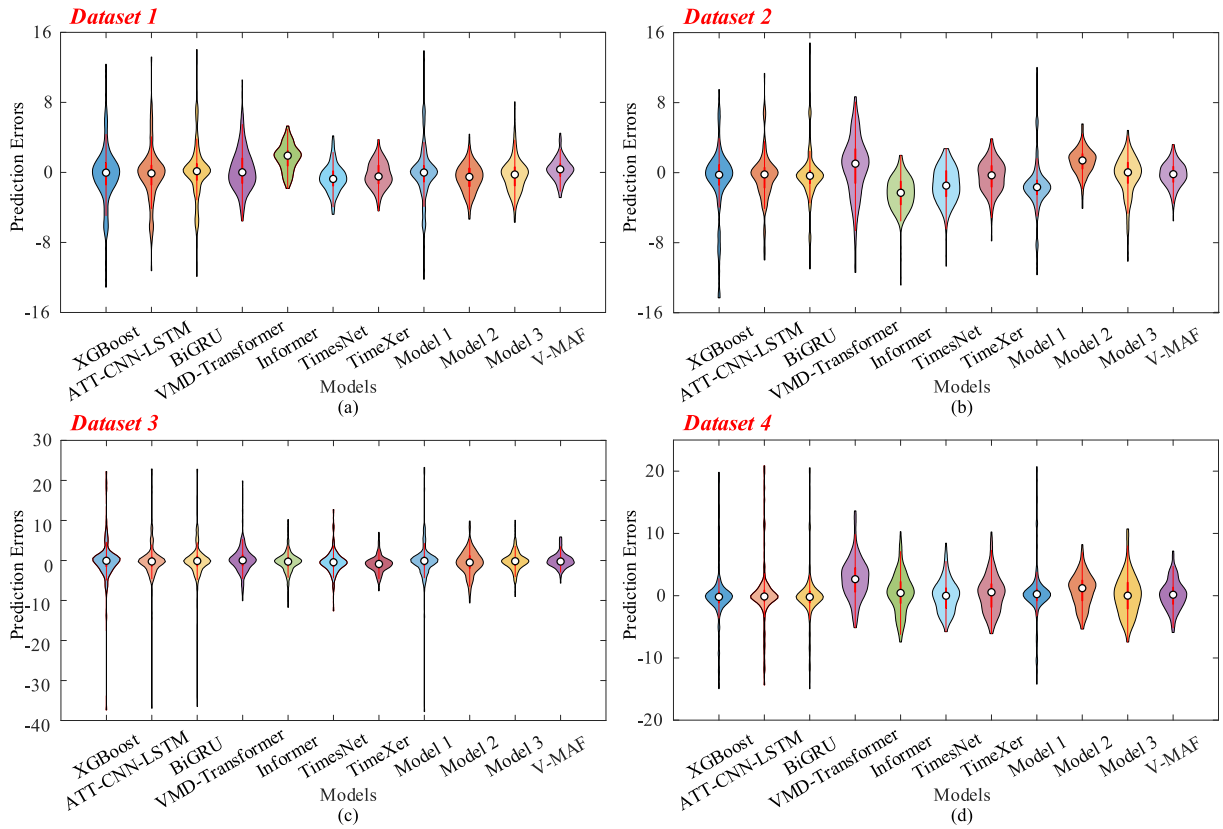

Fig. 12. Distribution of prediction errors of each model.

XGBoost, ATT-CNN-LSTM, BiGRU, and VMD-Transformer. Even when evaluated against high-performing models like Informer, TimesNet, and TimeXer, V-MAF maintains a slight yet consistent advantage. This comparative analysis clearly validates the proposed V-MAF forecasting method's superior accuracy and reliability.

4.3. Model stability analysis

The dispersion of the error set in the model's prediction results reflects the differences between the predicted and actual values, which can be used to assess the model's stability. This paper collected datasets of discrepancies between the prediction results of different models and the actual results, and conducted the dispersion analysis.

First, to intuitively understand the error distribution of each model, violin plots are used to visualize the shape, distribution, and central tendency of the prediction errors for each model across the datasets, as shown in Fig. 12.

According to Fig. 12, based on the analysis of the four datasets, the V-MAF model exhibits a narrower and more symmetric distribution of prediction errors, indicating its superior performance in error control, stability, and accuracy. The Informer, TimesNet, and TimeXer models follow, showing acceptable performance in some datasets but slightly inferior overall performance compared to the V-MAF model. Specifically, in Dataset 1, the Informer demonstrates a relatively narrow error distribution but with a positive bias, while in Dataset 2, it shows a higher lower bound of errors, resulting in unsatisfactory performance. The TimesNet performs relatively better in Dataset 1 compared to the Informer, with a smaller median deviation; however, in Datasets 2 and 4, its error distribution covers a wider range. The TimeXer exhibits a narrow error distribution in Datasets 1 and 3, but in Datasets 2 and 4, its error dispersion is greater than that of the V-MAF model. Model 2 and Model 3 show moderate performance, each with distinct characteristics across different datasets, yet their performance still lags behind that of the V-MAF model. Meanwhile, the XGBoost, ATT-CNN-LSTM, BIGRU, VMD-Transformer, and Model 1 models all display a high degree of error dispersion across the four datasets.

To quantitatively evaluate the dispersion of error values and the stability of the models, this study introduces two quantitative metrics: Standard Deviation (SD) and Mean Absolute Deviation (MAD). The formulas for these metrics are as follows:

$$\mathrm{SD} = \sqrt{\frac{\sum_{i=1}^{n} (e(i) - \overline{e})^2}{n}} \tag{19}$$

$$\mathrm{MAD} = \frac{\sum_{i=1}^{n} |e(i) - \overline{e}|}{n} \tag{20}$$

where $e(i)$ represents the error value at time i, and $\overline{e}$ represents the mean of the errors.

Table 10 presents the Quartiles (Q1, Q2, Q3), the SD, and the MAD of the error datasets for each model across the datasets. To ensure a clearer comparison, the models with the best SD and MAD values in each dataset are highlighted in bold, while the second-best models are underlined.

Based on the results from the four datasets, the V-MAF model demonstrates exceptional performance in terms of SD and MAD metrics. Particularly in Dataset 1 and Dataset 3, it achieves the lowest values for both SD and MAD, while in the other datasets, it ranks at least second-best in at least one of these metrics. This indicates that the V-MAF model exhibits smaller fluctuations in prediction error and higher stability. Additionally, the V-MAF model performs exceptionally well in terms of quartiles, especially in Dataset 1 and Dataset 2, where its prediction error distribution is more concentrated, further validating the model's stability. The TimeXer and TimesNet models also show strong performance in SD and MAD metrics across multiple datasets, particularly in Dataset 1 and Dataset 3, where their error fluctuations are

Table 10
Different model prediction error analysis.

Dataset	Prediction model	Metrics				
		Q1 (25 %)	Q2 (50 %)	Q3 (75 %)	SD	MAD
Dataset 1	Model 1	−1.0878	−0.0048	0.7852	3.1332	0.9399
	Model 2	−1.5922	−0.5098	0.4790	1.5904	1.0249
	Model 3	−1.5281	−0.2317	0.6240	1.9234	1.0635
	XGBoost	−1.4249	−0.0209	1.0988	3.2089	1.2788
	ATT-CNN-LSTM	−1.4279	−0.1018	0.8343	3.1822	1.0576
	BiGRU	−0.8810	0.1464	1.0054	3.1737	0.9614
	VMD-Transformer	−1.2406	0.0153	1.6376	2.4386	1.3971
	Informer	0.7453	1.9211	2.5034	1.5207	0.9609
	TimesNet	−1.5354	−0.7375	0.1746	1.5730	0.8935
	TimeXer	−1.2868	−0.4559	0.8197	1.4524	0.9755
	V-MAF	−0.4268	0.3618	0.9918	**1.2773**	**0.7045**
Dataset 2	Model 1	−2.5073	−1.6539	−0.8437	2.8197	**0.8195**
	Model 2	0.4286	1.4033	2.1627	**1.4887**	0.8360
	Model 3	−1.1990	0.0416	1.1775	2.2600	1.1892
	XGBoost	−1.4208	−0.2391	0.9127	4.0333	1.1537
	ATT-CNN-LSTM	−1.7468	−0.1915	0.6100	2.8832	1.0772
	BiGRU	−1.2245	−0.3514	0.2726	2.9848	0.8379
	VMD-Transformer	−1.2095	1.0348	2.7274	3.4569	1.8206
	Informer	−3.6649	−2.2913	−0.9723	1.9639	1.3274
	TimesNet	−2.7283	−1.4736	0.2209	2.0096	1.4960
	TimeXer	−1.6376	−0.2967	0.8066	1.7594	1.2590
	V-MAF	−1.0813	−0.1658	0.7065	1.5315	0.8984
Dataset 3	Model 1	−1.2652	−0.1001	0.9502	5.2928	1.0882
	Model 2	−1.9412	−0.4928	1.1999	3.0913	1.5337
	Model 3	−1.2280	−0.1784	0.7166	2.3646	1.0161
	XGBoost	−1.4977	−0.0808	0.9462	5.2496	1.3183
	ATT-CNN-LSTM	−1.4554	−0.2497	0.8591	5.2261	1.1107
	BiGRU	−1.3698	−0.1329	1.0900	5.2330	1.2229
	VMD-Transformer	−1.3530	0.0291	1.3798	3.3944	1.3507
	Informer	−1.3625	−0.2814	0.6395	2.5648	0.9878
	TimesNet	−1.5748	−0.4378	0.5705	2.7326	1.0683
	TimeXer	−1.8428	−0.8706	0.0451	2.0461	0.9579
	V-MAF	−1.0922	−0.2605	0.7876	**1.8466**	**0.9238**
Dataset 4	Model 1	−0.8018	0.2464	1.0584	4.5211	0.9360
	Model 2	−0.8042	1.2072	2.4896	2.6101	1.6662
	Model 3	−2.1176	0.0057	2.1432	3.2974	2.1234
	XGBoost	−1.1509	−0.1941	0.5889	4.5522	0.8908
	ATT-CNN-LSTM	−0.8833	−0.1252	0.8879	4.5680	**0.8401**
	BiGRU	−1.1438	−0.2135	0.7354	4.5701	0.9441
	VMD-Transformer	0.5933	2.6554	4.4727	3.3139	1.8873
	Informer	−1.2392	0.4388	2.2559	3.1960	1.7672
	TimesNet	−1.9979	−0.0090	1.2517	2.5989	1.6430
	TimeXer	−1.8573	0.5558	1.9063	2.9838	2.0253
	V-MAF	−1.3662	0.1595	1.3705	**2.3852**	1.4047

smaller, indicating greater stability. In contrast, Model 1 and Model 2 only achieve lower MAD and SD values in Dataset 2 but perform poorly across all four datasets, indicating insufficient overall stability. The performance of ATT-CNN-LSTM and XGBoost is similar to that of Model 1 and Model 2, while models such as BiGRU and VMD-Transformer perform poorly across all datasets, lacking the stability of the V-MAF model. Overall, the V-MAF model exhibits the best overall performance across multiple datasets, particularly excelling in error fluctuation control and stability.

5. Conclusion

The imbalance between electricity supply and demand leads to price fluctuations, underscoring the importance of accurate price forecasting for ensuring efficient energy utilization and market stability. This paper proposes a novel EPF method based on the V-MAF model. The approach is divided into three parts: data pre-processing, parallel network feature extraction, and feature fusion and prediction. Firstly, the PCC and MIC between past and current samples are computed. The results demonstrate that utilizing historical electricity price and load sequences to predict current electricity prices is feasible. Furthermore, the V-MAF model is designed. The model takes into account the volatility and noise interference inherent in electricity price and load data, as well as their potential multi-time-scale characteristics. Initially, VMD is employed to decompose the electricity price and load signals. Subsequently, the decomposed data are fed into a feature extraction module consisting of parallel SE-TCN and SE-GRU for multi-channel data feature exploration. The integrated features from both components are then fused using a MHA mechanism, enabling the model to focus on different time-scale components of the input features from various perspectives. Finally, prediction is performed through a dense layer. The effectiveness of this model has been validated using four datasets from real-world energy markets. The experimental results demonstrate that the V-MAF model is the best-performing model overall and is suitable for a variety of datasets. Taking Dataset 1 as an example, the details are as follows:

1) In the ablation experiments, the four metrics evaluating the model's prediction accuracy—RMSE, MAE, MAPE, and R_2—are 1.3168, 0.9995, 1.4248, and 0.9787, respectively, all of which represent the best performance. This confirms the contributions of each component in the proposed V-MAF model.
2) In the comparative experiments, the RMSE of the V-MAF model is reduced by 59.13 %, 58.75 %, 58.53 %, 46.36 %, 42.34 %, 22.65 %, and 11.09 % relative to XGBoost, ATT-CNN-LSTM, BiGRU, VMD-Transformer, Informer, TimesNet, and TimeXer, respectively.
3) The lowest values of SD (1.2773) and MAD (0.7045) further demonstrate that the stability of the proposed model is superior to that of other comparative models.

Overall, the V-MAF model proposed in this study exhibits the smallest prediction errors, high stability, and superior generalization capability compared to other models.

CRediT authorship contribution statement

Yuzhen Xu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Xin Huang:** Writing – review & editing, Writing – original draft, Visualization, Software, Formal analysis, Data curation, Conceptualization. **Ziao Gao:** Writing – review & editing, Writing – original draft, Visualization, Validation, Formal analysis, Data curation, Conceptualization. **Mohamed A. Mohamed:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tao Jin:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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